

WILL MORONG

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SUMMARY

Software Engineer with expertise building large-scale distributed systems and cloud infrastructure at Amazon, processing data for millions of customers. Proven track record leading cross-organizational initiatives including EU accessibility compliance, multi-region infrastructure migrations, and system reliability improvements. Experienced with ML-powered recommendation systems, AWS SageMaker, and applied NLP. Strong technical foundation in distributed computation (Apache Spark, Java/Scala), AWS services (EMR, EC2, S3, CloudFormation), and full-stack development with on-call operations experience.

PROFESSIONAL EXPERIENCE

Amazon — Software Development Engineer (L4)

Nov 2024 — Jan 2026

Personalization (SDO), Team: Zeitgeist | Seattle, WA

- Designed and maintained large-scale Apache Spark data pipelines in Java/Scala on AWS EMR, analyzing customer shopping behavior and integrating with ML-powered recommendation model APIs to surface personalized recommendations for millions of customers
- Leveraged AWS SageMaker for model inference integration within data pipelines, bridging ML outputs with production recommendation systems
- Developed customer-facing frontend experiences using TypeScript and CSS within Amazon's Cards and Kata frameworks, ensuring responsive design and optimal performance
- Served in monthly on-call rotation, debugging distributed systems issues and resolving high-severity production incidents while coordinating with partner teams across Personalization and retail organizations
- Collaborated with cross-functional teams on data ingestion pipelines, analytics infrastructure, and system availability improvements

Key Accomplishments:

EU Accessibility Compliance — Lead Engineer

- Sole engineer leading Zeitgeist's initiative to achieve European Accessibility Act (EAA) compliance across customer-facing UI, eliminating regulatory risk across 27 EU member states ahead of the 2025 deadline
- Implemented comprehensive accessible technology features: screen-reader compatibility, ARIA labels, keyboard navigation, focus management, and responsive layouts conforming to WCAG 2.1 AA standards
- Improved page load performance, reduced cumulative layout shift, and delivered measurably better usability for all customers including those with disabilities

Regional Infrastructure Migration — Lead Engineer

- Re-architected Personalization's Shared Fleet infrastructure to migrate compute workloads from capacity-constrained legacy AWS regions (DUB, PDX, IAD) to scalable new regions (ZAZ, FRA)
- Improved service resiliency, regional redundancy, and peak-traffic handling capacity through infrastructure-as-code updates (CloudFormation, YAML, JSON)
- Authored migration playbooks and coordinated rollout with platform engineering teams, enabling seamless traffic distribution

Multi-Architecture Deployment for Spark Clusters

- Built unified multi-architecture JAR for database-writing tool containing Rust shared libraries compiled for both amd64 and aarch64 architectures
- Implemented JNI runtime selection logic, enabling seamless deployment across heterogeneous EMR clusters and eliminating architecture-specific failures
- Reduced operational complexity and improved deployment reliability across diverse compute environments

JDK 17 Migration for Critical API Service

- Led migration of ZeitgeistDataService from JDK 8 to JDK 17, modernizing the API that serves all best-seller and new-releases data for Amazon's entire website
- Upgraded critical infrastructure serving 300 million customers monthly, improving security, performance, and long-term maintainability
- Coordinated testing and deployment across multiple environments, ensuring zero downtime for high-traffic customer-facing endpoints

ADDITIONAL EXPERIENCE

UNH InterOperability Lab — Datacenter Engineer Intern / Team Lead

Sept 2021 — May 2024

Durham, NH

- Led team of 4 engineers, mentoring new hires and managing laboratory infrastructure for networking hardware interoperability testing
- Architected and deployed KVM/QEMU/Libvirt virtualization stack, achieving approximately 4× performance improvement over legacy VMware infrastructure
- Migrated test automation platform from Java to Python, implementing multi-platform CI/CD pipelines and improving test reliability
- Developed systemd-boot customizations enabling NVMe boot on unsupported hardware and authored technical documentation

PROJECTS

Stuffchat — Self-Hosted Chat & Voice Platform

Personal Project | github.com/willmorong/stuffchat

- Designed and built a full-stack self-hosted team chat application from scratch — Rust/Actix backend, vanilla JS/HTML/CSS frontend, and a published native iOS app (App Store) — serving real users with channels, voice calls, file sharing, and message search
- Architected a standalone iOS push notification relay service that provisions APNS tokens against self-hosted servers, enabling native push notifications without a centralized cloud dependency
- Implemented real-time features including WebRTC-based voice and screen sharing, WebSocket messaging, full-text search (SQLite FTS5), custom emoji, and a role/permissions system with bitfield capability flags
- Shipped iteratively to the [App Store](#) (peaked at #27 in Social Apps (Paid) on the US Charts) and server repo (v1.12, 196 commits), managing the full software lifecycle: design, implementation, testing, deployment, and ongoing maintenance

Fine-Tuned LLM Discord Bot

Personal Project

- Self-hosted a LLaMA 8B model and fine-tuned it on a curated dataset of a friend's Discord messages to generate stylistically accurate, contextual responses for comedic effect
- Managed the full ML pipeline: data collection and preprocessing, fine-tuning with LoRA adapters, local inference serving, and Discord bot integration via Python
- Gained hands-on experience with NLP fine-tuning techniques, transformer architectures, and deploying LLMs in resource-constrained environments (the model was quantized and made to run on an 8GB RX 5700XT using Vulkan Compute API)

TECHNICAL SKILLS

Languages: Java, Scala, Python, TypeScript, JavaScript, Rust, C/C++

Machine Learning & AI: LLM fine-tuning (LoRA/QLoRA), NLP, AWS SageMaker, recommendation model APIs, transformer architectures, PyTorch

Big Data & Distributed Systems: Apache Spark, AWS EMR, EMRFS, distributed computation, data pipeline design

Cloud & Infrastructure: AWS (EC2, S3, CloudFormation, IAM), CI/CD pipelines, Linux administration, containerization, KVM/QEMU virtualization

Frontend: TypeScript, CSS, HTML5, web accessibility (WCAG, ARIA), responsive UI design

Tools: Git, Bitbucket, JIRA, Confluence

EDUCATION

University of New Hampshire

Durham, NH

Bachelor of Science in Computer Science

2020 — 2024

Relevant Coursework: Algorithms, Operating Systems, Cybersecurity, Natural Language Processing, Object-Oriented Software Development